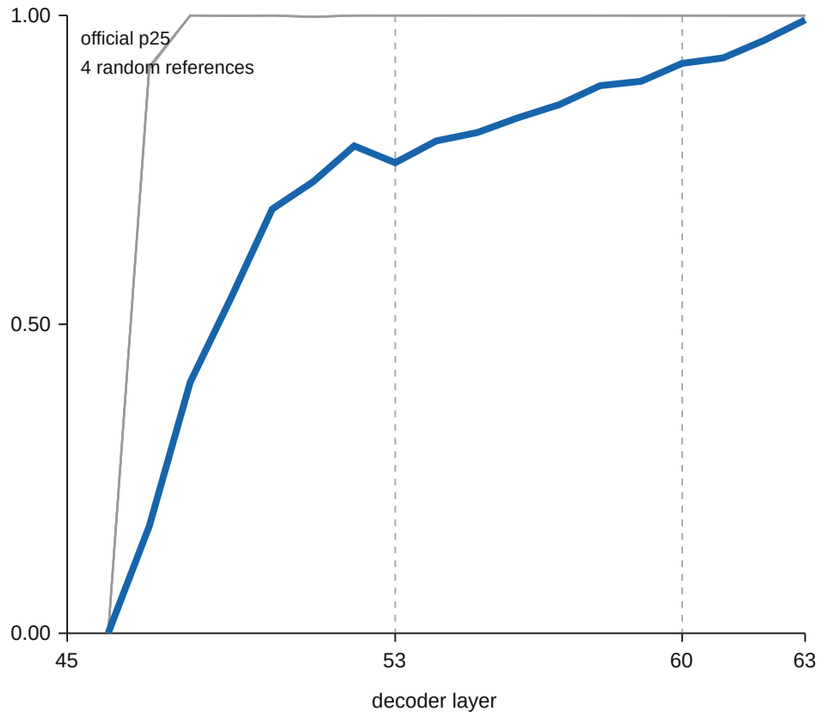
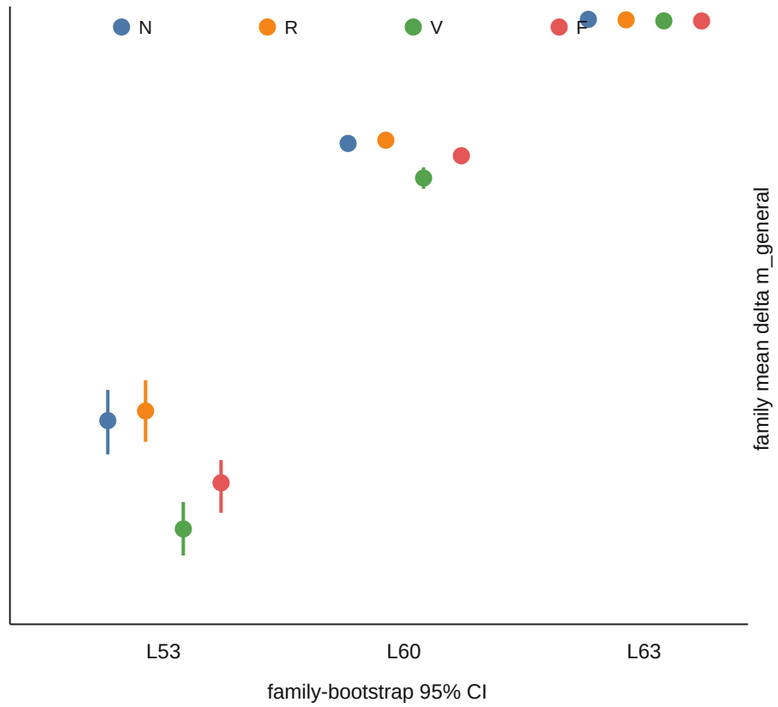
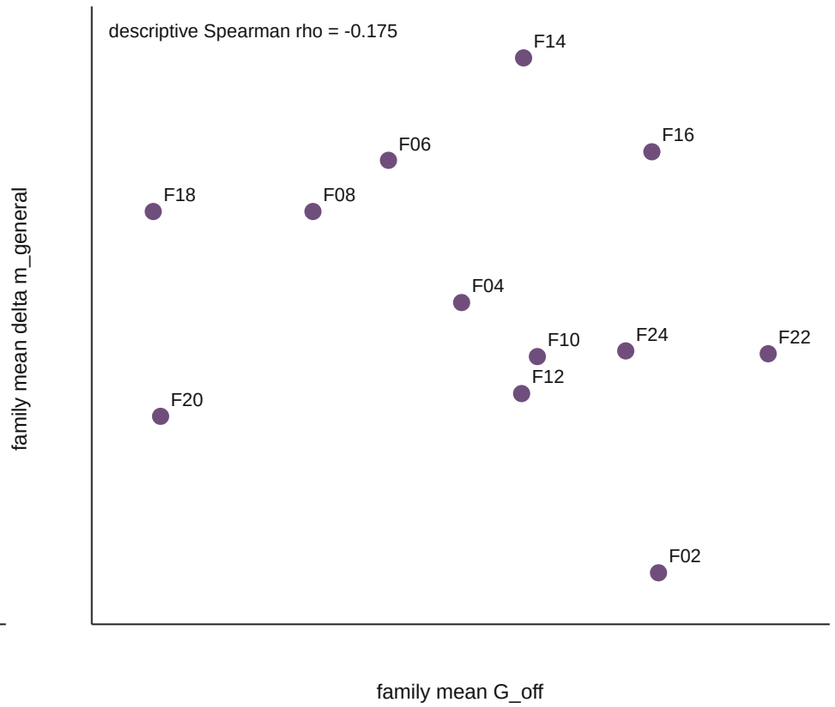
A Downstream off-direction share



B Arm breakdown

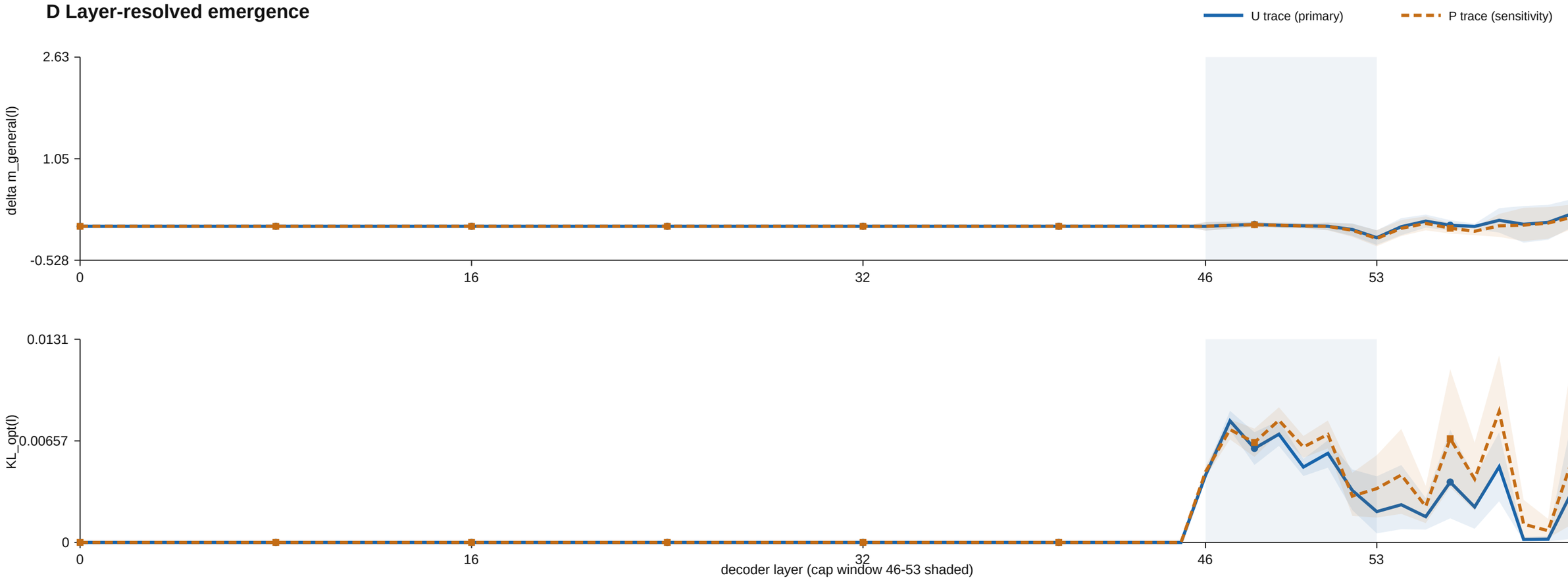


C Descriptive family-level association



Downstream projections use the fixed layer-53 cap direction, not a layer-specific Assistant Axis; random references use their corresponding signed r53. Four references are a direction battery, not a random null distribution. Panel C is descriptive and has no fitted line.

D Layer-resolved emergence



cache-faithful replay; divergence below layer 46 reflects cache-propagated effects, not direct rectification. Logit-lens readout; descriptive.